An explicit reconstruction lattice for a Néron-normalized seed theta kernel

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Abstract

For the fixed rank-two curve of the v0.2.2 dossier, we record a source-reviewed written deduction that the Néron-normalized kernel of an integral degree-zero theta packet lies in $(2^{10} \cdot 4547)^{-1}\mathbf{Z}$. A comparison of congruence ideals shows that its scale relative to the primitive quaternionic eigenvector is a unit at every good prime $p \geq 5$. These statements give a finite integer acceptance test for the retained good-prime witness without computing the unknown quaternionic norm. No seed kernel value or new primary component is computed. The imported theorems and retained geometric inputs are explicit; this supplement has model reviews, no independent human review and no new Lean verification. It does not change the v0.2.2 manuscript or its claim grades.

1 Objects and fixed normalizations

The curve and conductor are

$$E : y^2 = x^3 - 22735x^2 + 106272484x,$$
$$N = 2^3 \cdot 3 \cdot 2243 \cdot 4547^2 \cdot 5843.$$

The translation $x = X + 7578$ has $u = 1$ and gives the retained global minimal model

$$[0, -1, 0, -66020924, -65074271436].$$

Thus $\omega_E = dx/(2y)$ is the Néron differential. Existing claims A.1, A.3 and A.4 supply the local data, full odd residual images and the chosen-map Manin bound respectively.

Let $N^- = 3 \cdot 2243 \cdot 5843$ and $N^+ = 2^3 \cdot 4547^2$. In the definite quaternion algebra ramified at N^- and infinity, use the retained Eichler order of level N^+ and its ideal-class set X. The stabilizer weights are all one. Let $\psi \in \mathbf{Z}[X]$ be the primitive vector on the one-dimensional f_E-line and put $M = \sum_{x \in X} \psi(x)^2$. For $z = \sum z_x[x]$ write $\psi(z) = \sum z_x \psi(x)$ and $\deg z = \sum z_x$.

The retained theta convention is

$$\Theta_{x,y}(\tau) = \frac{1}{2} \sum_{b \in L(x,y)} e^{2\pi i Q(b)\tau}, \quad \Theta(z, w) = \sum_{x,y} z_x w_y \Theta_{x,y}.$$

Here $L(x, y)$ is the normalized Hom lattice, even of rank four, with Gram determinant N^2 and level N. With $B(1) = \mathrm{Id}$, the audited prime-to-level Hecke projection satisfies

$$\lambda_f(\Theta(z, w)) = \frac{\psi(z)\psi(w)}{M}. \tag{1}$$

This is the retained rank-one theta identity, not a global identification of M with a classical congruence number or modular degree.

Let E_* be the optimal quotient, $\phi_0 : X_0(N) \to E_*$ its parametrization of degree d, and $h : E_* \to E$ the identity or the retained rational 2-isogeny. Set $e = \deg h \in \{1, 2\}$ and $\phi_E = h\phi_0$, with

$$\phi_E^* \omega_E = c_E \omega_f, \qquad \omega_f = 2\pi i f(\tau)\, d\tau, \qquad c_E \mid 2^9.$$

The Manin bound refers to this chosen map, without further multiplication maps. Its retained geometric dependencies are not reproved here. Define

$$A_E = \frac{i}{2} \int_{E(\mathbf{C})} \omega_E \wedge \overline{\omega_E}, \qquad P_f = \int_{\Gamma_0(N)\backslash \mathfrak{H}} |f|^2 \, dx\, dy.$$

A_E is the complex period-lattice area, not the BSD real period or the Néron–Tate regulator. Pullback of area gives

$$ed\, A_E = 4\pi^2 c_E^2 P_f, \qquad \mathscr{K}_E(G) := \frac{4\pi^2}{A_E} \langle G, f \rangle = \frac{ed}{c_E^2} \lambda_f(G). \tag{2}$$

Consequently $\mathscr{K}_E(z, w) := \mathscr{K}_E(\Theta(z, w)) = u_E \psi(z)\psi(w)$, where $u_E = ed/(c_E^2 M)$.

2 The good-prime unit comparison (NK.1)

Theorem 1 (NK.1; written deduction with published imports)**.** *For every prime $p \geq 5$ with $p \nmid N$,*

$$v_p(M) = v_p(d), \qquad u_E \in \mathbf{Z}_p^\times.$$

In particular, for $z \in \mathbf{Z}[X]$, $\mathscr{K}_E(z, z)$ is a p-adic unit if and only if $\psi(z) \not\equiv 0 \pmod p$.

Proof. We spell out the comparison chain, with the source hypotheses retained. In the weight-two pairing of Kim–Ota [1, §6.1, (6.1)], the Atkin–Lehner element permutes X and acts on its f-line by a sign. All stabilizer weights are one. The pairing matrix on $\mathbf{Z}_p^X$ is therefore a permutation matrix and is perfect. The globally primitive vector is locally primitive, so the image of the pairing on its eigenline is $M\mathbf{Z}_p$ (the sign is a unit). This identifies the quaternionic pairing ideal ξ_{f_B}, not yet a classical congruence ideal.

The seed satisfies Kim–Ota's CR+ hypotheses. The weight is two and $p \geq 5$ is prime to N. The residual image is $\mathrm{GL}_2(\mathbf{F}_p)$; its restriction to $\mathbf{Q}(\sqrt{p^*})$ contains $\mathrm{SL}_2(\mathbf{F}_p)$ and is absolutely irreducible. The integer N^- is squarefree with three prime factors, and no prime exactly divides N^+. At $3, 2243, 5843$ the minimal discriminant valuations $2, 1, 2$ are prime to p, so the residual Tate inertia remains nontrivial. At 2 and 4547 inertia is finite of order dividing 24 and of order 4 respectively. Its averaging idempotents are p-integral, so invariant dimensions and Artin conductors are unchanged on reduction. Thus $N(\bar\rho) = N$, and all the Tamagawa exponents in [1, Corollary 6.7] vanish. That corollary gives

$$v_p(M) = v_p(\xi_{f_B}) = v_p(\eta_f(N^+, N^-)) = v_p(\eta_f(N)).$$

At nonordinary primes the freeness used here explicitly imports [1, Remark 6.6(1)], the low-weight crystalline adaptation. Its deformation-theoretic justification is not reproved in this supplement.

For completeness, the last classical lattice identification uses the *full* Hecke algebra $\mathbb{T}$ and the standard duality $S_2(\Gamma_0(N), \mathbf{Z}) = \mathrm{Hom}_{\mathbf{Z}}(\mathbb{T}, \mathbf{Z})$. If e_f is the rational idempotent, write $\mathbb{T} \cap \mathbf{Q} e_f = \mathbf{Z}(ne_f)$ with ne_f primitive. Then the annihilator definition gives $\eta_f(N) = n\mathbf{Z}$ locally, while duality

shows that the projection of integral cuspforms is $n^{-1}\mathbf{Z}f$. The quotient by $\mathbf{Z}f$ and its orthogonal complement is cyclic of order n. Hence $v_p(\eta_f(N)) = v_p(r)$ for the classical congruence number r. The full-Hecke comparison and the localization are detailed in the retained Phase 1 proof; they are not inferred solely from (1).

Agashe–Ribet–Stein [2, Theorem 2.1] give $v_p(r) = v_p(d)$ for the optimal elliptic quotient at good p. Finally $e \in \{1, 2\}$ and $c_E \mid 2^9$ imply the asserted unit property of u_E. □

The conclusion is equality of valuations at the stated primes. It asserts neither $p \nmid M$ nor a global equality $M = r$ or $M = d$.

3 The explicit real reconstruction lattice (NK.2)

Lemma 1. *For $z, w \in \mathbf{Z}[X]$ with $\deg z = 0$, $G = \Theta(z, w)$ is cuspidal at every cusp and integral at infinity. At a cusp denominator $c \mid N$, set $n = v_q(N)$ and $a = v_q(c)$. Every positive Fourier coefficient in the plain weight-two slash expansion has valuation at least $a - n$.*

Proof. We use the retained Poisson expansion (lab Lemmas 4 and 4′). For $\gamma = \left(\begin{smallmatrix} A & B \\ c & D \end{smallmatrix}\right)$, $c \geq 1$, write $\mathrm{e}(t) = e^{2\pi i t}$. The expansion is

$$(\Theta_{x,y}|_2\gamma)(\tau) = -\frac{1}{2Nc^2} \sum_{\ell \in L(x,y)^\#} S(A, \ell; c)\, \mathrm{e}(DQ(\ell)/c)\, \mathrm{e}(Q(\ell)\tau),$$

$$S(A, \ell; c) = \sum_{r \in L(x,y)/cL(x,y)} \mathrm{e}\big((AQ(r) + (\ell, r))/c\big).$$

There is no additional cusp-width factor in the slash operator. The zero-vector Gauss value is independent of (x, y), including its local norm-unit choices. Every constant term of G is therefore a common scalar times $\deg z \deg w$ and vanishes. At infinity the positive coefficients of each $\Theta_{x,y}$ are integers by pairing b and $-b$.

At an Eichler prime the two off-diagonal summations force their support conditions and contribute q^{2a}. The remaining hyperbolic sum contributes q^a times a root of unity. Thus the local Gauss value is zero or q^{3a} times a root of unity. At a division prime, $n = 1$; the retained norm Gauss identity gives $-q^3$ times a root of unity on support when $a = 1$, and the factor is 1 when $a = 0$. Pairing ℓ and $-\ell$ cancels the theta factor $1/2$ exactly, including at $q = 2$. The prefactor has valuation $-n - 2a$, and other local factors are q-units or zero. Hence each surviving summand, and its finite sum at a positive Fourier frequency, has valuation at least $a - n$. This is a lower bound, not a statement of integrality at every cusp. □

Theorem 2 (NK.2; written deduction with retained Manin input)**.** *For $z, w \in \mathbf{Z}[X]$, $\deg z = 0$, put $G = \Theta(z, w)$. Then*

$$D_0\omega_G \text{ lies in the Néron cotangent lattice of } J_0(N), \qquad D_0 = 2 \cdot 4547,$$

and

$$\boxed{B_\theta \mathscr{K}_E(G) \in \mathbf{Z}, \qquad B_\theta = 2^{10} \cdot 4547 = 4\,656\,128.}$$

The weaker multiplier $B_{\mathrm{join}} = 2^{17} \cdot 4547 = 595\,984\,384$ also works.

Proof. CNS [3, Proposition 5.14], which applies to general cusp forms in its stated domain, requires the respective valuation bounds

$$-n \quad (a=0), \qquad a-n+\frac{1}{q-1} \quad (0<a<n), \qquad 0 \quad (a=n).$$

Lemma 1 meets these bounds at $q \in \{3, 2243, 5843\}$, where $n=1$. At 2 ($n=3$) and 4547 ($n=2$), multiplying once by the local prime meets all intermediate bounds. At good primes integrality at infinity suffices. Thus $D_0\omega_G$ is a regular differential. These are sufficient multiplier exponents; their optimality for the actual packet forms is not established.

CNS [3, Theorem 6.12] gives the required rational singularities at every prime: at all primes at least five; at three since $v_3(N)=1$; and at two since $3 \mid N$ and $3 \equiv 3 \pmod 4$. Proposition 6.2 and Example 6.3 of that paper identify regular differentials with the Néron cotangent lattice. Local integrality at all primes gives the asserted global lattice membership.

Let $\pi_E : J_0(N) \to E$ be the chosen quotient and $i_E : E \to J_0(N)$ its dual, so $\pi_E i_E = [ed]$. Hecke equivariance kills the complement of the f-line under i_E^*, while $i_E^*\pi_E^*\omega_E = ed\omega_E$. Therefore

$$i_E^*\omega_G = \frac{ed}{c_E}\lambda_f(G)\omega_E = c_E\mathscr{K}_E(G)\omega_E.$$

The map extends to Néron models, and ω_E generates the integral cotangent lattice of E. It follows that $D_0 c_E \mathscr{K}_E(G) \in \mathbf{Z}$. The retained $c_E \mid 512$ gives B_θ. The weaker multiplier follows also by using the retained whole-lattice inclusion $2^8 S \subseteq H$ at two. This latter argument avoids the theta Gauss estimate at two; both arguments retain the same Manin bound. □

4 A certificate specification, not a computed witness (NK.3)

For an admissible good prime $p \geq 5$, $p \nmid 7N$, use the retained BLV conductor-p^2 Gross packet. If γ generates its first cyclic layer, its mod-p weight is

$$\overline{z}_p = \sum_{j=0}^{p-1} j^2[\gamma^j P_1], \qquad M_2(p) = \bar{\psi}(\overline{z}_p).$$

Here the Gross point P_1 is not the Mordell–Weil basis point bearing that name in the main paper. Equal trace degrees and $\sum_{j\in\mathbf{F}_p} j^2 = 0$ make the mod-p degree zero. An integral lift z_p need not have degree zero: balanced square weights sum to 22 at $p=11$. Choose a degree-one class $[a]$ and set

$$z_p^0 = z_p - \deg(z_p)[a]. \tag{3}$$

Then $\deg z_p^0 = 0$ and $z_p^0 \equiv \overline{z}_p \pmod p$. The integral weights and anchor must be specified in a certificate.

Corollary 1 (NK.3; conditional sufficient criterion)**.** *For this lift define $n_p = B_\theta \mathscr{K}_E(z_p^0, z_p^0)$. Then*

$$n_p \in \mathbf{Z}, \qquad p \nmid n_p \iff M_2(p) \neq 0 \implies \mathrm{Sha}(E/\mathbf{Q})[p^\infty] = 0.$$

The last implication imports the existing ordinary/supersingular forward bridge B.5; its antecedent is uncomputed here.

Proof. Theorem 2 gives the integer, and $p \nmid B_\theta$. By Theorem 1, $n_p = B_\theta u_E \psi(z_p^0)^2$ has unit scale. Equation (3) identifies its nonvanishing test with $M_2(p)$. Apply B.5. No converse from vanishing of Sha to the witness is used. □

Let $[L, U]$ be a genuinely certified real interval for the *final* normalized scalar, including all numerical and truncation errors. Unique integer recovery is possible when its radius is strictly less than

$$\frac{1}{2B_\theta} = \frac{1}{9\,312\,256}; \qquad \text{fallback: } \frac{1}{2B_{\text{join}}} = \frac{1}{1\,191\,968\,768}.$$

The bound guarantees at most one lattice point; the theorem and enclosure supply existence. For a wider interval put $k_- = \lceil B_\theta L \rceil$ and $k_+ = \lfloor B_\theta U \rfloor$. A sufficient acceptance test is

$$k_- \le k_+, \qquad \left\lceil \frac{k_-}{p} \right\rceil > \left\lfloor \frac{k_+}{p} \right\rfloor.$$

The first inequality rejects an empty lattice interval. The second excludes every multiple of p. A nonunit or unresolved interval gives no conclusion about Sha. The sharp lattice cannot be applied to a noncuspidal anchor diagonal merely because an anchored expression eventually produces this cuspidal scalar.

5 Evidence grade and remaining obligations

The detailed proof sources, source hashes and three model-review records are included with this supplement. The portable checks replay 262 exact Eichler Gauss identities, 13 cusp-threshold comparisons and degree-zero relifting, and compare 41,925 interval decisions with direct integer enumeration. They verify finite algebra and acceptance arithmetic; they do not formalize the source papers or evaluate a seed numerator. The general division-place Gauss identity, the chosen-map Manin bound, and Kim–Ota's nonordinary adaptation retain their stated written/import grades. Numerical period checks are evidence only. There is no claim of independent human review or of novelty relative to the literature.

The reconstruction prerequisite is now supplied for this normalized degree-zero route. An affordable certified seed evaluation and a nonzero seed witness are still missing. The fixed denominator B_θ is not a fixed numerator controlling the prime support of Sha: n_p varies with p and its packet. No new primary component, FIN, ORDER9 or Q36 follows. Full BSD still requires an all-prime arithmetic conclusion and the exact real leading-term comparison.

References

[1] C.-H. Kim and K. Ota, *On the quantitative variation of congruence ideals and integral periods of modular forms*, source version arXiv:1905.02926v4, §6.1, Assumption 6.3, Theorem 6.5, Remark 6.6(1), Corollary 6.7.

[2] A. Agashe, K. Ribet and W. Stein, *The modular degree, congruence primes, and multiplicity one*, author-hosted manuscript, Theorems 2.1 and 3.6, Remark 3.7.

[3] K. Česnavičius, M. Neururer and A. Saha, *The Manin constant and the modular degree*, source version arXiv:1911.09446v3, Propositions 5.14 and 6.2, Example 6.3, Theorem 6.12.

[4] BSD seed lab, source commit `b358f7507759dad4a1e016c519d6f486ffca37e1`, *01_NORMALIZATION_REVIEW.md* and *02_DENOMINATOR_LATTICE.md*, 24 September 2026. Copied with their context, source bindings and scoped reviews in this supplement.